The Hecke Algebra of a Pair (G, H)

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1 Setup

Let G be a finite group and H a subgroup. The cosets G/H form a finite set of size $n = [G : H]$. The group G acts on G/H by left multiplication: g sends the coset xH to gxH. This is a permutation action — each $g \in G$ permutes the n cosets. Linearizing, we get a representation of G on the vector space $\mathbb{C}[G/H]$ with basis $\{\delta_{xH} : xH \in G/H\}$, where g acts by the permutation matrix P_g that sends δ_{xH} to δ_{gxH}.

2 The orbital graphs

The double cosets $H\backslash G/H$ partition G. Each double coset HgH determines a graph on the vertex set G/H: the Sabidussi graph $\mathrm{Sab}(G, H, HgH)$, with an edge from xH to yH whenever $xy^{-1} \in HgH$.

Since the double cosets partition G, these graphs partition the edges of the complete directed graph on G/H: every ordered pair of cosets is an edge in exactly one orbital graph.

The trivial double coset $HeH = H$ gives the diagonal orbital — a loop at every vertex.

3 The adjacency matrices

Each orbital graph has an adjacency matrix T_{HgH} of size $n \times n$. The (xH, yH) entry is

$$(T_{HgH})_{xH,\,yH} = \begin{cases} 1 & \text{if } xy^{-1} \in HgH, \\ 0 & \text{otherwise.} \end{cases}$$

Equivalently, T_{HgH} maps each coset to the sum of its neighbours in the orbital graph:

$$T_{HgH}(\delta_{xH}) = \sum_{\substack{yH \in G/H \\ xy^{-1} \in HgH}} \delta_{yH}.$$

The identity double coset $H = HeH$ gives the identity matrix.

4 These matrices are G-equivariant

Each T_{HgH} commutes with the G-action on $\mathbb{C}[G/H]$. That is, for every $z \in G$,

$$T_{HgH}(z \cdot \delta_{xH}) = z \cdot T_{HgH}(\delta_{xH}).$$

This holds because the orbital graph is G-symmetric: G acts on the graph by automorphisms, so it commutes with the adjacency matrix.

5 The orbital adjacency matrices form a basis for $\mathrm{End}_G(\mathbb{C}[G/H])$

Proposition 1. *The set* $\{T_{HgH} : HgH \in H\backslash G/H\}$ *is a basis for* $\mathrm{End}_G(\mathbb{C}[G/H])$.

Proof. A G-endomorphism T of $\mathbb{C}[G/H]$ is a linear map $T\colon \mathbb{C}[G/H] \to \mathbb{C}[G/H]$ satisfying $T(g \cdot v) = g \cdot T(v)$ for all $g \in G$. A linear map on an n-dimensional space has n^2 parameters. We will show that equivariance cuts this to n, and then H-invariance cuts n down to $r = |H\backslash G/H|$.

Step 1: $n^2 \to n$. Since $yH = y \cdot eH$, equivariance gives $T(\delta_{yH}) = y \cdot T(\delta_{eH})$ for any coset yH. So T is completely determined by the single vector

$$T(\delta_{eH}) = \sum_{xH \in G/H} a_{xH}\, \delta_{xH},$$

a list of n coefficients a_{xH}, one per coset.

Step 2: $n \to r$. For any $h \in H$, applying Step 1 with $y = h$ gives $T(\delta_{hH}) = h \cdot T(\delta_{eH})$. But $hH = eH$, so the left side is $T(\delta_{eH})$. Therefore

$$\sum_{xH} a_{xH}\, \delta_{xH} = h \cdot \sum_{xH} a_{xH}\, \delta_{xH} = \sum_{xH} a_{xH}\, \delta_{hxH}.$$

Comparing coefficients: $a_{hxH} = a_{xH}$ for every $h \in H$ and every coset xH. This is the orbit relation for the action of H on G/H by left multiplication: two cosets xH and yH must get the same coefficient whenever they lie in the same H-orbit, i.e., whenever $y \in Hx$. The H-orbit of xH is $\{hxH : h \in H\}$, and the union of these left cosets in G is the double coset HxH. (The right-hand H plays no role here: xH and $xh'H$ are already the same coset for any $h' \in H$.)

So the n coefficients collapse to $r = |H\backslash G/H|$ free parameters, one per double coset.

Conversely, every such coefficient pattern is realised: given any choice of scalars a_d for $d \in H\backslash G/H$, define $T(\delta_{eH}) = \sum_d a_d \sum_{xH \subseteq d} \delta_{xH}$ and extend by equivariance. This is well-defined precisely because the coefficients are H-invariant. The result is:

$$T = \sum_{d \in H\backslash G/H} a_d \, T_{HdH}.$$

Every G-endomorphism is a linear combination of the T_{HgH}, and every such linear combination is a G-endomorphism.

Independence. The double cosets partition G, so the orbital graphs partition the edges of the complete directed graph. Their adjacency matrices have 1s in disjoint positions. No non-trivial linear combination can be zero. □

6 The Hecke algebra

The space $\mathrm{End}_G(\mathbb{C}[G/H])$ is closed under composition: if S and T both commute with the G-action, then so does ST, since for any $g \in G$,

$$(ST)(g \cdot v) = S(g \cdot T(v)) = g \cdot S(T(v)) = g \cdot (ST)(v).$$

Therefore the product of two basis elements decomposes as a linear combination of basis elements:

$$T_{Hg_1H} \cdot T_{Hg_2H} = \sum_d c_d \, T_{HdH}.$$

The structure constant c_d is a non-negative integer: it counts the number of two-step paths from eH to any fixed coset in HdH, taking one edge in $\mathrm{Sab}(G, H, Hg_2H)$ followed by one edge in $\mathrm{Sab}(G, H, Hg_1H)$.

The algebra $\mathrm{End}_G(\mathbb{C}[G/H])$ with this multiplication is the *Hecke algebra* $\mathcal{H}(G,H)$. Equivalently, it is the commutant of the permutation matrices $\{P_g : g \in G\}$ inside $\mathrm{Mat}_n(\mathbb{C})$, where $n = |G/H|$. Here each P_g is the $n \times n$ permutation matrix for the action of g on the cosets G/H, and the commutant is the set of all $n \times n$ matrices that commute with every P_g.